

CERTIFICATE

This is to certify that the project report entitled "**FocusFlow: A MERN-Stack Productivity Ecosystem for Modern Learners**" submitted by **Amanul Haque** (Roll No: CS-2341697) to IILM University, Greater Noida in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my supervision.

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ABSTRACT

In the modern educational landscape, digital distractions, frequent context-switching, and procrastination heavily impact student productivity. Existing task management applications often suffer from fragmented workflows, requiring users to juggle separate apps for to-do lists, timers, and note-taking. This project introduces **FocusFlow**, a comprehensive MERN-stack (MongoDB, Express.js, React, Node.js) web application designed specifically as a cohesive productivity ecosystem for modern learners.

FocusFlow addresses the gaps in traditional tools by integrating Intelligent Task Management, an embedded Pomodoro timer, Contextual Notes, and a unique "Recovery Mode" for overcoming creative blocks and burnout. The system utilizes Natural Language Processing (NLP) for quick task entry, allowing users to create scheduled tasks conversationally. A dynamic priority algorithm categorizes tasks automatically based on the Eisenhower matrix principles. Furthermore, FocusFlow visualizes user productivity through interactive charts mapping "Focus Hours" and task completion rates. The application ensures high performance, secure JWT authentication, and a responsive UI built with Tailwind CSS. Experimental results and user testing demonstrate that FocusFlow significantly reduces app-switching and improves time-management adherence among students.

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